



Hubble is back!

The telescope which was offline since mid-June is now online thanks to the backup hardware. <https://digit.in/aug21-37>



RNA editing to increase yield

Manipulating RNA can allow plants to yield dramatically more crops. <https://digit.in/aug21-38>



HUNTING FOR SUPERSYMMETRY

Exploring a dark frontier of physics

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hen the Higgs boson was spotted in 2012, all the particles in the Standard Model had been hunted

down, which describes all the particles that we know of. Physicists suspect there is physics beyond the Standard Model, with more particles to be discovered. There are some questions such as the nature of dark matter, or why there is more matter than antimatter, or why neutrinos have masses or something known as a hierarchy problem, which are stats for the universe that seem unbalanced. The weak force, for example, is much stronger than gravity, or gravity is just weaker than the three other fundamental forces, and researchers believe that there might

be other physics that provide good explanations for all of these questions. The Standard Model is inconsistent with general relativity, and there are some phenomena beyond explanation by both of them such as singularities and the Big Bang. The Standard Model predicted gluons and some additional bosons and quarks, but the particle accelerators around the world are hoping to see glimpses of other particles, ones predicted by other models.

The Higgs boson that we have found is referred to as the vanilla Higgs or the Standard Model Higgs, and even the SM like Higgs, as physicists believe there is at least one other, heavier Higgs boson still waiting to be discovered. This is because the one we discovered is just too light. There is an elegant approach to extend the Standard Model, known as supersymmetry, which predicts superpartners for every particle in the standard model, with slightly different properties. These are called squarks, sleptons, gauginos and higgsinos. The

superpartners of the Higgs boson are called higgsinos, and the experiments at the LHC are in search of these higgsinos. The thing is that physicists have been searching for these superpartners of standard model particles for years at the LHC, and have not found any. Considering the energy levels at which the LHC operates at, the physicists should have found plenty of particles beyond those predicted by the standard model, and especially the ones predicted by the many supersymmetric models.

There are plenty of possible explanations. The superpartners to the existing particles may be too heavy to be detected. Another possibility is that they are hiding in plain sight. If all or many of the supersymmetric particles are close in mass range to the lightest supersymmetric particle (the LSP), then they would all be very difficult to detect. As all the masses of these supersymmetric particles are “compressed” together, these are difficult to pry apart and study. If the superpartner to the Higgs boson,